

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 4th Year 2nd Term Examination-2021
LE 4025
Materials Science for Footwear Products

Time: 3.0 Hours

Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION-A

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|------|--|----|
| 1(a) | Classify shoe upper materials. | 05 |
| 1(b) | “Poromeric is known as man-made leather”- Why? Explain poromeric materials and its layers. | 10 |
| 1(c) | Explore the benefits and drawbacks of using leather against synthetic materials for shoe uppers. | 10 |
| 1(d) | What is meant by adherent? State the principles of adhesion. | 10 |
| 2(a) | Define the following adhesive terminologies:
(i) Green strength (ii) Spotting tack (iii) Peel strength and (iv) Shelf life. | 16 |
| 2(b) | List out the ingredients of PU sole with their functions. | 09 |
| 2(c) | “Hot melt adhesive can vary depending on application temperature, application method and ingredients”- Explain it. | 10 |
| 3(a) | Analyze the factors that influence the adhesion strength between the material and the adhesive surfaces. | 10 |
| 3(b) | Why do rubber soles require vulcanization? Write down the flow chart of producing vulcanized rubber soles in different methods. | 09 |
| 3(c) | How can you derive flexibility of a polymer material from “Mark-Houwink” equation? | 08 |
| 3(d) | “The shoe soles must be dimensional stable”-state the reason. Briefly explain the factors that influence in dimensional stability of shoe soles. | 08 |
| 4(a) | How to carry out PVC polymerization for shoe sole? Mention the advantages and disadvantages of PVC soling materials. | 14 |
| 4(b) | “Thermoplastic Rubber (TPR) soling materials have two phase structure”- Explain the reason. | 08 |
| 4(c) | What are the functions of plasticizer and filler in producing TRR sole? | 05 |
| 4(d) | Describe the sequential operations briefly for producing Ethylene Vinyl Acetate (EVA) foam sheets. | 08 |

SECTION-B

5(a)	Define toe-puff. Discuss the following toe-puff materials: (i) Paint on Puff, (ii) Celluloid impregnated fabric and (iii) Leather.	11
5(b)	State the manufacturing and wearing requirements of an insole forepart and seat board.	10
5(c)	Discuss compound insole. What are the limitations of wood shank?	08
5(d)	Draw different parts of a heel.	06
6(a)	Write short notes on stacked heel. How can you replace the impression of stacked heel with heel cover?	10
6(b)	If the shank length is 50 mm, then calculate total shank depth. Explain austempering process.	09
6(c)	Draw different parts of an eyelet. Differentiate between eyelet and grommet.	10
6(d)	Sketch parts of a buckle with proper indications.	06
7(a)	Define shoe finishing. What are the objectives of shoe finishing?	05
7(b)	Explain the following shoe lacing method with figures: (i) Over under lacing-Even number of eyelets and (ii) Shoe shop lacing.	10
7(c)	Define aglet. Calculate the shoelace length of a typical sneaker with horizontal spacing of 50 mm and 5 pairs of eyelet.	05
7(d)	Name the type, use and melting point of the following waxes: (i) Carnauba and (ii) Monton. Why carnauba wax is superior to others?	10
7(e)	Show the proper position of reinforcement in strap with figures.	05
8(a)	Draw a flow chart on shoe polish manufacturing.	07
8(b)	Explain the ingredients of a black boot polish.	08
8(c)	Write short notes on (i) overall reinforcement, (ii) eyelet reinforcement. Provide figures.	10
8(d)	Discuss trims attachment by stitching method and two-part fittings with necessary figures.	10

Department of Leather Engineering
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B. Sc. Engineering 4th Year 2nd Term Examination-2021
LE 4213
Testing of Leather and Allied Materials

Time: 3.0 Hours

Full Marks: 210

- N.B. i) Answer any **THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION-A

- 1 Mr. Barry Allen, newly appointed in a testing company, visited a leather industry few days ago. He and his team were asked to inspect the "Full leather" samples from a vehicle loaded with 379 samples. The team had a doubt regarding the resiliency of the upper leather samples. After a discussion, they picked some samples for laboratory testing and sent these to the lab. Now answer the following questions:
- 1(a) (i) Write a full report on selecting the samples for thumb tests as well as sketch a table indicating all the tests to be satisfied (✓) or unsatisfied (×). 14
- 1(b) (ii) Write down the procedure that Mr. Allen and his team followed to inspect the resiliency of the samples and how this problem can be controlled? 08
- 1(c) (iii) State the sample selection Mr. Allen did for the lab test, sampling position (sketch) and conditioning that needed to be done before conducting the tests. 13
- 2 Cisco Ramon, one of Barry's friends, works in the laboratory of the same testing company. Mr. Roman conducted the tests like tensile strength, elongation at break, all the tearing strengths. In case of ball bursting test, the grain crack strength of leather of 0.9 mm thickness had been determined with a lastometer 7735.81 lb/ft at a distance 7 mm. He also noticed that at load 8 kg, the distension was 3 mm.
- (i) How light fastness of finish films and water fastness properties of leather can be enhanced? 11
- (ii) Write down the factors that Mr. Ramon kept in mind for tensile strength of leather. 08
- (iii) "To inspect the overall strength of leather, lastometer test is conducted"- prove this statement with necessary sketches. 08
- (iv) "Stitch and tongue tear strength can be expressed by a straight line equation"- explain with graph. 08
- 3 Mr. Harrison Wells conducted a tensile strength test of a leather specimen of thickness 1.2 mm. The load was recorded 6345 lb. The jaws of the tensile tester machine was observed to move from 3 cm to 8 cm.
- (i) Calculate the tensile strength and percentage of elongation and compare with the standard values. (1 kg=2.204 lb) 11
- (ii) "Leather is intermediate between rubber and plastic"-prove mathematically with graph. 08
- (iii) Explain the relation between load and distension following Brodeley's equation of 08

lastometer tester.	
(iv) Show the mechanism of leather specimen tearing.	08
4(a) Describe the controlling factors of finish film bondage.	09
4(b) State the significance of moisture, pH and chromic oxide content test of leather.	08
4(c) Why several conditionings are required for the test of resistance to ageing?	08
4(d) Explain the "Amos and Thompson" model of bond strength test.	10

SECTION-B

5(a) Mention the effects of fogging. State the fogging test in reflectometric method for automotive leather	16
5(b) Briefly explain the factors determining the fogging behavior of upholstery leather.	09
5(c) Write short notes on: (i) Corrosion test and (ii) Lace to eyelet test.	10
6(a) Describe the principle, conditions, inspection stages and standard values of Vamp Flexing test.	15
6(b) Mention the purpose of different dia circular disc in case of softness test determination.	07
6(c) Define Fick's law of diffusion. If the water vapour permeability (WVP) of an upper leather is found 0.0832 gms/min/sq.cm at the end of first 30 min and 0.078 gms/min/sq.cm at the end of second 30 min. If the diameter of the circular opening of the cap of the bottle is half of 100 mm, calculate the weight of the bottle system at different stages when the final weight is 148.6 gms.	08
6(d) Shrinkage temperature of leather can be increased more than 120°C-explain how?	05
7(a) State the principle and testing procedure of abrasion resistance tested soling materials in the rotating drum method.	15
7(b) Briefly explain the peel strength of sole attachment with footwear. Mention the standard value of peel strength during sole adhesion.	07
7(c) State specifically the methodology and inspection stages of Ross flexing test.	06
7(d) How can you increase and decrease the Flexing endurance property of upper leather.	07
8(a) Define scuffing. How can you increase the scuff resistance of upper leather?	07
8(b) How perspiration damages leather and how to improve perspiration resistance of leather?	12
8(c) Write the principle of the following tests (i) Ross Flexing test and (ii) Water Proofness test	10
8(d) What are fasteners and accessories? Why these are used during shoe and leather goods manufacturing?	06

Department of Leather Engineering
Khulna University of Engineering & Technology
B. Sc. Engineering 4th Year 2nd Term Examination-2021
IPE 4219
Industrial Management & Entrepreneurship

Time: 3.0 Hours

Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION-A

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|------|--|----|
| 1(a) | Define the term "Equity". Discuss briefly various managerial skills with suitable example(s). | 13 |
| 1(b) | Write down the merits and demerits of classical management theory. | 12 |
| 1(c) | Explain contingency theory school briefly. | 10 |
| 2(a) | Demonstrate the three types of superior subordinate relationship identified by management expert Gsaicunas. | 13 |
| 2(b) | Briefly explain the obstacles to effective delegation process. | 12 |
| 2(c) | Write down the advantages of line and staff organization structure with necessary figures. | 10 |
| 3(a) | Differentiate between ERG theory of motivation and achievement theory of motivation. | 12 |
| 3(b) | Describe the effect of conflict on organization performance. | 09 |
| 3(c) | A job is rated in terms of wages Tk. 25 per hour. The standard time set for the job is 12 hours. The worker has taken 8 hours for the completion of the job. Calculate the workers wage rate per hour using Halsey plan and Rowan plan. Also, find out the efficiency of the worker and mention the incentives to be received by the worker. | 14 |
| 4(a) | Define human resource planning. Briefly explain various types of involuntary separation. | 11 |
| 4(b) | Briefly discuss the counseling process. | 10 |
| 4(c) | Describe point system method and factor comparison method of job evaluation. | 14 |

SECTION-B

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|------|--|----|
| 5(a) | Define entrepreneurship. Differentiate between entrepreneurship and management. | 13 |
| 5(b) | What is key performance Indicators (KIPs)? | 12 |
| 5(c) | Discuss about different characteristics of Management Information System (MIS). | 10 |
| 6(a) | What is Limited Liability Partnership? What are the differences among company, partnership and LLP? Discuss. | 15 |

6(b)	What do you understand by oligarchic management in respect of company?	08
6(c)	Briefly explain about five taxonomies for Decision Support System (DSS).	12
7(a)	Differentiate between top down and bottom up budgeting.	10
7(b)	What is strategic management? Briefly explain about SWOT analysis.	12
7(c)	Why job rotation is done? What are the potential disadvantages of job rotation?	13
8(a)	Define technology transfer. Discuss about various modes of technology transfer.	13
8(b)	Why most of the new entrepreneurships fail? What steps are needed to solve these problems?	12
8(c)	What do you understand by technology entrepreneurship? Briefly discuss about technology developers and technology users.	10

Department of Leather Engineering
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 B. Sc. Engineering 4th Year 2nd Term Examination-2021
LE 4207
Dyeing & Finishing II

Time: 3.0 Hours

Full Marks: 210

- N.B. i) Answer any **THREE** questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

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|---|----|
| 1(a) What is an ideal solution? Explain the non-ideal behavior of dye solution. | 08 |
| 1(b) How do electrolytes play a vital role in leather dyeing? | 07 |
| 1(c) How the dyeing process comes about? | 08 |
| 1(d) Describe the forces of attraction between dyes and leather fibers. | 12 |
| 2(a) Determine the dye aggregation number by the porous plate diffusion method. | 14 |
| 2(b) Estimating molecular size from diffusion fibrinogen has a diffusion co-efficient of about $2.0 \times 10^{-7} \text{ cm}^2/\text{sec}$ at 37°C . It is believe to be rod-shaped, about thirty times longer than it is wide. How large is the molecule? The viscosity of fibrinogen is $0.00695 \text{ g cm}^{-1}\text{sec}^{-1}$. | 10 |
| 2(c) Explain the factors affecting diffusibility. | 11 |
| 3(a) Dyeing equilibrium has greater impact in leather dyeing justify it. | 15 |
| 3(b) Deifne the drawbacks of Fick's first law of diffusion. | 10 |
| 3(c) Explain the importance of half time of dyeing in leather dyeing. | 10 |
| 4(a) Write short notes on: (i) Dyeing equilibrium (ii) Adsorption isotherm (iii) Affinity of dyes and (iv) Entropy of dyeing. | 12 |
| 4(b) How can you find out the dyeing equilibrium? | 08 |
| 4(c) Write the mathematical form of Freundlich, Nernst and Langmuir adsorption isotherm. Explain the different symbols used in the equation. | 07 |
| 4(d) Explain the mechanisms of diffusion with the figures. | 08 |

SECTION-B

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|--|----|
| 5(a) Why following terms should be considered during leather dyeing? (i) Reflection (ii) Refraction and (iii) Dispersion | 05 |
| 5(b) How can we perceive secondary and tertiary color although we have dedicated cone cells for primary colors? | 10 |
| 5(c) Identify the different limitations of trichromatic theory and deduce their probable remedy. | 10 |
| 5(d) We cannot see "reddish green" or "bluish bellow"- Justify the statement. | 10 |
| 6(a) What is the importance of Kubelka-Munk equation in leather coating? | 10 |

6(b)	State the limitations of Oswald color system.	10
6(c)	By using CIE color system determine the RGB primaries of following lights reflected from colored leather surface: (i) 490 nm (ii) 520 nm and (iii) 480 nm. Also identify the position of these wavelengths on the diagram.	15
7(a)	Different light sources directly influence the appearance of colored leather surface- Justify.	05
7(b)	How does color temperature affect our life?	10
7(c)	How will you determine the performance of leather coating by assessing: (i) Resistance to bleeding and (ii) Resistance to dressing agent	10
7(d)	Rheology of coating is associated with leather dyeing and finishing—Justify the statement.	10
8(a)	Briefly explain the flow problems and associated solutions of low VOC cutting technologies.	12
8(b)	Write down the effect of weathering on leather coating.	08
8(c)	Write down the characteristics of acrylic and polyurethane coating.	08
8(d)	What factors can ensure the flexible leather coating than viscosity liquid?	07

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 B. Sc. Engineering 4th Year 2nd Term Examination-2021
LE 4217
Production & Quality Control

Time: 3.0 Hours

Full Marks: 210

- N.B. i) Answer any **THREE** questions from each section in separate scripts.
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 iii) Assume reasonable data if any missing.

SECTION-A

- 1(a) State the law of variable proportion. Briefly discuss the stages of production based on the law of variable proportion with suitable example. 15
- 1(b) Write down the factors affecting the selection of production process. Which production system should be followed as an entrepreneur to establish a small leather good industry near Fulbarigate with initial production target of 1000 pieces of men's wallet per month with 02 varieties? State the methods with its advantages and disadvantages. 12
- 1(c) Define productivity. Point out the key steps for improving productivity. 08
- 2(a) What is planning and control? Briefly discuss the production planning and control process with a flow diagram. 10
- 2(b) Actual demands of a footwear company for the last 10 years are shown in Table. 18
- (i) Calculate demand forecast for the year of 2017 using three periods weighted moving average method.
- (ii) Calculate exponential smoothing forecast for the year of 2022 assuming forecast for the year of 2017 using $\alpha = 0.2$, and
- (iii) Calculate the MAD error of the exponential smoothing forecast for the period of 2017-2021.

Table			
Year	Actual demand	Year	Actual demand
2012	1600	2017	2300
2013	1800	2018	2100
2014	2000	2019	1900
2015	1600	2020	2200
2016	2200	2021	2400

- 2(c) Distinguish between open and closed loop system of production control. 07
- 3(a) What is forecast? "Forecasting is the art and science of predicting future events" – Explain. 06
- 3(b) Briefly discuss the factors determining production planning procedures. 09
- 3(c) The demand forecast for a footwear industry are shown in the Table-(i) for the time horizon January to April. The beginning inventory and labour are 400 units and 40 workers, respectively. Now make an aggregate plan of the company according to chase and level strategy and find out the best on strategy. The relevant costs are given in Table-(ii). 20

Table (i)					
	Jan.	Feb.	Mar.	April	Total
Demand forecast	1800	1500	1100	900	5300
Working days	22	19	21	22	84

Table (ii)			
Inventory holding cost	\$ 2.0/U/M	Labour time required	0.20 U/hr.
Back orders	\$ 5.0 /U/M	Straight time cost (8 hrs.)	\$ 5/hr.
Hiring & Training cost	\$ 250/worker	Outsourcing cost	\$25/U
Lay off cost	\$ 400/worker		

4(a)	Define aggregate planning of production. What are the points available for the managers to meet the fluctuating future demand?	10
4(b)	How routing, scheduling and loading are inter-connected?	05
4(c)	What is material handling? Briefly discuss the general considerations for selecting material handling equipment.	12
4(d)	Develop a plant layout of a footwear industry for improving productivity.	08

SECTION-B

5(a)	What is Total Quality Management (TQM)? As a leather engineer, how can you implement TQM in a footwear industry?	08
5(b)	Write down the differences between quality control and quality assurance.	08
5(c)	What is six sigma? State the aim and concept of six sigma for leather industry.	09
5(d)	Why zero defects concept is necessary to maintain the quality of an industry? How can you implement the principle of zero defects?	10
6(a)	State the objectives and key elements of TQM for the implementation in footwear industry.	09
6(b)	Make the differentiation between TQM and six sigma.	08
6(c)	What is Environmental Management System (EMS)? Write down the objectives of EMS.	09
6(d)	Define Environmental Auditing (EA). Why EA is necessary to deliver the best quality products and maintain the goodwill of an organization.	09
7(a)	What is Environmental Impact Assessment (EIA)? Write down the goals and phases of EIA.	09
7(b)	Assume, a footwear company estimates its carrying cost at 25% and ordering cost at \$ 5 per order. The estimate annual requirement is 50,000 units at a price of \$ 3 per unit. Calculate the: (i) Economic Order Quantity (EOQ) (ii) Number of order to be placed in a year.	08
7(c)	What is inventory management? State the factors that influence the inventory management.	09
7(d)	What is inventory control? Write down the function of inventory control.	09
8(a)	Why scheduling and sequencing are important for leather industry?	08
8(b)	Describe the benefits of Gantt chart for a footwear industry?	09
8(c)	Discuss the principle of ISO 14000 series at a glance.	09
8(d)	Define quality. Write down the concept of quality for leather products.	09